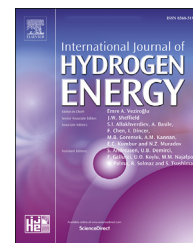


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Effect of inoculum pre-treatment on mesophilic hydrogen and methane production from food waste using two-stage anaerobic digestion

Razieh Rafieenia^a, Alberto Pivato^{b,*}, Maria Cristina Lavagnolo^b

^a Department of Industrial Engineering, University of Padova, Via Marzolo No.9, 35131, Padova, Italy

^b Department of Civil, Environmental and Architectural Engineering, University of Padova, Via Marzolo No.9, 35131, Padova, Italy

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ABSTRACT

Two-stage anaerobic digestion of food waste was performed using four different inoculum pre-treatment methods to enrich hydrogen (H_2) producing bacteria from sludge. The pre-treatments used in this study included heat shock, alkaline treatment, aeration, and a novel pretreatment using waste frying oil (WFO). Alkaline pretreatment and aeration did not completely inhibit methanogens in the first stage while no methane (CH_4) was detected in the reactors cultivated either with heat shock or WFO-pretreated inocula. The highest H_2 and CH_4 yields (76.1 and 598.2 mL/gVS, respectively) were obtained using the inoculum pretreated with WFO. The highest total energy yield (21.96 kJ/gVS) and total organic carbon (TOC) removal efficiencies (95.77%) were obtained using inoculum pretreatment with WFO. The total energy yield trend obtained using the different pretreatments was as follows: WFO > alkaline > heat > aeration > control.

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Introduction

Hydrogen (H_2) has been regarded as a sustainable energy carrier and termed “fuel for the future” due to its carbon free nature that does not contribute to the emission of greenhouse gases when combusted. Furthermore, its energy yield (142 kJ/g) is 2.75 times greater than that of all carbon fuels [1]. H_2 can be produced by anaerobic bacteria in a process called dark fermentation (DF) using inexpensive organic wastes as a substrate. Subsequent to DF, a major fraction of the organic matter remains in the liquid phase in the form of organic acids that should be treated before disposal. A promising system to address this issue could be the integration of dark

fermentation (DF) and anaerobic digestion (AD) which subsequently leads to waste reduction as well as enhanced bio-energy recovery from organic wastes. Two-stage AD has been proven as a biological treatment method for waste stabilization and energy recovery from a wide variety of organic wastes [2–7]. The possibility of combining DF and AD is currently receiving growing interest due to the further conversion of DF effluents, shorter substrate retention time, improved bio-stabilization of organic wastes, and higher energy yields compared to single-stage AD [8–10].

Full scale studies have shown that the addition of sludge collected from wastewater treatment plants could increase the amount of energy recoverable from organic wastes [11,12]. However, pre-treatment of the sludge prior to DF is an

* Corresponding author.

E-mail address: alberto.pivato@unipd.it (A. Pivato).

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indispensable step to inhibit H_2 consuming populations and improve H_2 production using anaerobic mixed cultures. Different methods have been used to enrich H_2 producing bacteria and suppress H_2 consuming species include heat shock [13,14], acid or alkaline pre-treatment [15,16], addition of chemical inhibitors [17–19], ultrasonication [20], gamma irradiation [21], and aeration [2,22].

The inhibitory effect of long chain fatty acids (LCFAs) on methanogens have been shown previously [17,18,23,24] where the addition of various LCFAs to anaerobic mixed culture to suppress methanogenic H_2 consumption was studied. All of these studies showed that the inhibitory levels of LCFAs for H_2 producing bacteria are significantly higher than that for methanogens and therefore controlling the LCFAs levels in the reactor could be used as a strategy to suppress methanogenic activity. However, due to the presence of LCFAs in the DF effluents, they cannot be used for the subsequent ad. Recently, Rafieenia et al. [19], reported the inhibitory effect of LCFAs on hydrogenotrophic methanogens even after removing the LCFAs after pretreatment. These results showed that anaerobic microbiota could be pretreated with waste frying oil (WFO) as a source of LCFAs and after pre-treatment, the soluble phase containing solubilised LCFAs might be removed. LCFAs can be adsorbed on the cell wall on specific microbial populations and inhibit their growth by interfering with nutrient transfer into the cell and also facilitating the accumulation of toxic substances inside the cell [25,26].

Selection of the best inoculum pre-treatment for H_2 production depends on substrate and inoculum type. There are several studies that investigated different inoculum pre-treatments for H_2 production from glucose [16–18,27,28]. Also, there are few reports that compared different inoculum pre-treatment methods for H_2 production from complex wastes including potato and pumpkin waste [22], corn stover [16], waste ground wheat [29], and stale corn [30]. However, to the best of the authors' knowledge, no other studies have reported the effects of inoculum pre-treatment methods on two-stage AD for H_2 and CH_4 production from food waste (FW). Most of the two-stage AD studies have been performed using heat shock as the inoculum pre-treatment [31]. Therefore, in order to fill this knowledge gap, this study aims to:

- 1) Investigate the possibility of using anaerobic mixed cultures pretreated with WFO in two-stage AD for H_2 and CH_4 production.
- 2) Compare the two-stage AD of FW using WFO pretreated cultures with three common inoculum pre-treatment methods (aeration, heat shock and alkaline pre-treatment).
- 3) Evaluate the overall performance of two-stage AD in terms of energy yield and substrate degradation.

Materials and methods

Seed sludge and pre-treatments

Granular sludge, used as the inoculum, was collected from a full-scale Up-flow Anaerobic Sludge Blanket (UASB) digester of a brewery factory located in Padova, Italy. Granular sludge

was characterised by a Total Solids (TS) concentration of 15% and Volatile Solids (VS) concentration of 47% TS. Four different pre-treatments were used to enrich H_2 producing bacteria:

- Heat shock: Granular sludge was boiled at 90 °C for 30 min [32].
- Aeration: The granular sludge was aerated for 24 h using an aquarium pump with an air flow rate of 3 L/min [15].
- Alkaline pre-treatment: The pH of the granular sludge was adjusted to 12.0 ± 0.1 with 3 N NaOH and maintained for 24 h [6].
- Pre-treatment with WFO: a saponified WFO solution was prepared according to the method described by Rafieenia et al., [19]. Fifteen g/L of WFO was added to the granular sludge cultures (33 gVS sludge/L saponified WFO solution) and maintained for 24 h. After the treatment, the pre-treated cultures were washed three times with tap water. In order to wash the granular sludge, the pretreated cultures were remained stagnant for 30 min to precipitate the granular sludge. After which, the supernatant containing saponified WFO solution was removed using a syringe. The washing was repeated two additional times by adding tap water and removing the supernatant.

Control cultures were also prepared without any form of pre-treatment.

It should be mentioned that pre-treatment conditions for each method might vary from one study to another and optimal conditions depend on both the inoculum and substrate. Therefore, the pre-treatment conditions applied in the present study were chosen from the studies that used similar inoculum or substrate.

Substrate

Synthetic FW samples were prepared to simulate the FW composition in Italy [33]. FW samples were mixed and shredded in a kitchen mill to make a homogenous blend and then analysed. The composition and characteristics of the FW are shown in Table 1.

Two-stage anaerobic digestion tests

Two-stage batch AD tests were performed using 500 mL glass reactors with a working volume of 250 mL. In the first stage (H_2 production), pretreated cultures as well as the control received 5 gVS/L FW. Followed by a pH adjustment to 5.5 using NaOH (3 M) and HCl (3 M) [4]. Subsequently, the bottles were sealed with silicone rubber stoppers, purged with nitrogen for 3 min to ensure anaerobic conditions were achieved, and incubated in a water bath at 35 °C. After 96 h (when the biogas production ceased), the reactors were opened and samples (5 mL) were taken from each reactor and stored in a freezer (–20 °C) for further analysis. Before storage in the freezer, the samples were filtered using membrane filters with a pore size of 0.45 μ m to be ready for dissolved organic carbon (DOC) and volatile fatty acid (VFA) analysis. In order to start the second stage (CH_4 production), 5 gVS/L of untreated granular sludge was added to the reactors [4]. The pH of all the reactors was

Table 1 – Characteristics of the inoculum and food waste used for two-stage AD.

Parameter	Food Waste	Inoculum
TS (%)	28.1	15.2
VS (% TS)	94.6	47.3
TOC (% TS)	49.7	28.6
TKN (% TS)	4.2	41.5
Meat (% W/W)	13.4	–
Cheese (% W/W)	5.5	–
Bread-pasta (% W/W)	10.8	–
Fruit (% W/W)	54.1	–
Vegetables (% W/W)	16.2	–
C/N	11.83	0.68

adjusted at 7.0 since it is the optimal pH for methanogens [34] and incubated at 35 °C. The second stage lasted for 32 days. All of the tests were performed in triplicate.

Analytical methods

Biogas composition was analysed using a gas-chromatograph (Varian 490-GC) equipped with a 10-m MS5A column to measure H₂ and CH₄ concentrations and a 10-m PPU column to analyse carbon dioxide (CO₂) concentration and two thermal conductivity detectors. Argon was used as the carrier gas with a pressure of 60 kPa in the columns. Temperatures of the column and injector were set to 80 °C. The volume of biogas produced during the two-stage AD tests was measured by means of the water displacement method. H₂, CO₂, and CH₄ volumes produced during the first and second stages of AD, were calculated according to Eq. (1) [35]:

$$V_{c,t} = C_{c,t} * V_{b,t} + V_H * (C_{c,t} - C_{c,t-1}) \quad (1)$$

where:

$V_{c,t}$ = Volume of H₂, CO₂, or CH₄ produced in the interval between t and $t - 1$.

$V_{b,t}$ = Volume of total biogas produced in the interval between t and $t - 1$.

V_H = Volume of the headspace of reactors.

$C_{c,t}$ and $C_{c,t-1}$ = Concentrations of H₂, CO₂, or CH₄ measured at times t and $t - 1$, respectively.

TS, VS, total organic carbon (TOC) and total Kjeldahl nitrogen (TKN) were analysed according to standard methods (APHA/AWWA/WEF, 2012). TOC content was measured using a TOC analyzer (TOC-V CSN, Shimadzu). In order to measure DOC concentrations, the effluent obtained at the end of the first and second stages of AD were filtered using membrane filters with a pore size of 0.45 µm before analysing using a TOC analyzer. In order to study the effects of various inoculum pre-treatments on VFA production, the effluents of the first stage were first filtered using membrane filters with a pore size of 0.45 µm. After which, the VFA concentrations was analysed using a gas chromatograph (Varian 3900) equipped with a CP-WAX 58 WCOT fused silica column and a Flame Ionization Detector. Nitrogen was used as the carrier gas with a flow rate of 4 mL/min in the column. The temperature of oven was set at 80 °C for 1 min and then increased to 180 °C (remained for 2 min) at a rate of 10 °C/min. The temperatures of the column and injector were held at 250 °C.

A modified Gompertz equation was used to model H₂ and CH₄ production and quantification of kinetic parameters (Eq. (2)). The modified Gompertz model is the most widely used method to analyse biogas production performance and therefore it was ideal to evaluate the H₂ and CH₄ productions using different inoculum pre-treatment methods.

$$H(t) = P * \exp \left\{ -\exp \left[\frac{R_m * e}{P} \right] (\lambda - t) + 1 \right\} \quad (2)$$

where:

H is the cumulative hydrogen production potential (ml).

P the H₂ production potential (ml).

R_m is the maximum hydrogen production rate (ml/h).

λ is the lag phase (h).

t is incubation time.

P , λ and R_m were estimated by using a curve fitting tool in Matlab (The Mathworks Inc., version 2016a).

Results and discussion

Effect of inoculum pre-treatment methods on the first stage of anaerobic digestion

Hydrogen production

In the first stage of AD, no detectable CH₄ was observed in the reactors pre-treated either with WFO or heat shock, indicating the efficiency of the mentioned pre-treatments on the inhibition of methanogens. For alkali treated and pre-aerated inoculum, small amounts of CH₄ were detected in the produced biogas (4.4 and 3.2%, respectively). Cumulative H₂ production obtained in the first stage of AD using various inoculum pre-treatment methods are shown in Fig. 1. H₂ produced by the control cultures was less than that of all pre-treatments (27.6 mL/gVS). The highest H₂ yield was obtained for WFO pre-treated cultures (76.1 mL/gVS) followed by heat shock (53.8 mL/gVS). H₂ yields of 42.8 mL H₂/gVS and 35.3 mL H₂/gVS were obtained for alkali pretreated and pre-aerated inoculum, respectively. Higher H₂ yields obtained by WFO-pre-treated inoculum compared to heat shock suggest that inoculum pre-treatment with WFO could provide a less severe condition for H₂ producing bacteria to be enriched. The lower H₂ yields obtained by aeration and alkaline pre-treatment

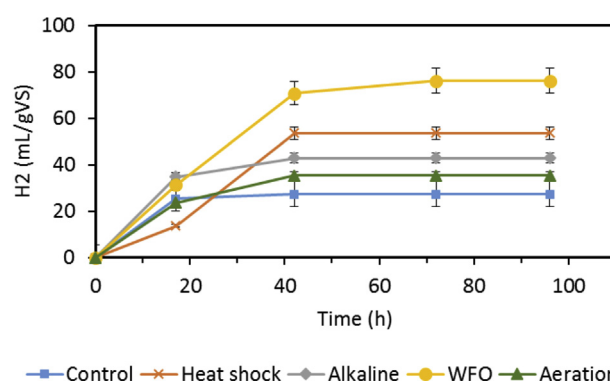


Fig. 1 – Cumulative H₂ production using different inoculum pre-treatment methods in the first stage of AD.

could be attributed to the incapability of the mentioned methods to suppress H_2 producing bacteria.

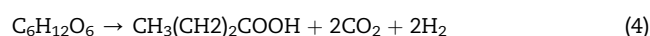
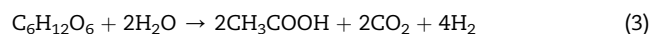
Kinetic parameters calculated using the modified Gompertz equation are shown in Table 2. The results showed that H_2 production increased for all of the pre-treatment methods when compared to the control. Inoculum pre-treatment with WFO and heat shock increased the cumulative H_2 production potential (P) by 182% and 109%, respectively, compared to the untreated culture. For alkaline and aeration pre-treatments cumulative H_2 productions were 56.5% and 31.2% higher than the control, respectively. Regarding the lag time (λ), the shortest lag time of 2.4 h was obtained for untreated inoculum while the longest lag time of 12.4 h was obtained for heat shock. The literature also reported that the longest lag time was observed after heat shock pre-treatment [37,38]. The highest lag time for heat shock pretreated cultures could be presumably due to the inhibition of some H_2 producing bacteria during the heat shock pre-treatment. Therefore, additional time is needed for H_2 producing bacteria to proliferate and produce H_2 .

Based on these results, the best inoculum pre-treatment method to enrich H_2 producing bacteria reported by a specific study might differ from other studies. This inconsistency could be due to the strong effect of substrate biodegradability on efficiency of each inoculum pre-treatment method. This means that the best inoculum pre-treatment using a specific substrate is not necessarily as effective using other substrates. Many studies have used different substrate pre-treatment methods prior to DF in order to increase the hydrolysis rate and nutrient conversion efficiencies [39–42]. However, the techno-economic feasibility of the combined processes needs to be evaluated for industrial application [43]. Moreover, composition variability of the inoculum is another issue that affects the specific inoculum pre-treatments and therefore results in contradiction between other studies. Microbial populations present in the initial inocula show different tolerability towards the changes in the environmental conditions and act differently to the severe conditions imposed by pre-treatments. The majority of the comparison studies on the inoculum pre-treatment methods have been performed using glucose as substrates. Comparison studies on H_2

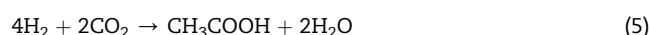
production from complex substrates using various inoculum pre-treatment methods are rare in the literature. Yin et al. [38], investigated H_2 production from glucose using four inoculum pre-treatment methods namely heat shock, gamma irradiation, acid, and alkaline pre-treatment and reported the highest H_2 yields for gamma-irradiated cultures. Using the same inoculum, a significant variation was observed in the H_2 production performances when glucose was replaced by raw grass as the lowest H_2 yield was obtained by gamma-irradiated cultures [44]. Ghimire et al. [22], evaluated H_2 production from potato and pumpkin waste using aeration, heat shock, and bromoethane sulphonate (BESA) addition and achieved the highest H_2 yield using BESA treatment. Dong et al. [45], reported that heat shock was the best inoculum pre-treatment to enhance H_2 yields from the organic fraction of municipal waste. Similarly, heat shock has been reported as the best inoculum pre-treatment using palm oil mill effluent [46], waste ground wheat [29], brewery wastewater [47], and corn stover hydrolysate [16].

Volatile fatty acids production

H_2 production through DF is accompanied with the production of soluble metabolites and mainly VFAs. VFAs produced in the first stage of AD using different inoculum pre-treatment methods are shown in Fig. 2. Acetic acid and butyric acid were the main VFAs detected by Gas Chromatography for all the pre-treatments. Lower acetic acid concentrations in the untreated culture could be attributed to the possible conversion of acetic acid to CH_4 by acetoclastic methanogens. The highest concentration of butyric acid was observed for WFO pre-treated inoculums (385.33 mg/L) while the alkali treated cultures produced the highest amount of acetic acid (258.66 mg/L). Several studies observed that H_2 yield was directly proportional to the butyrate/acetate ratio [48,49] while other studies reported no correlation between the butyrate/acetate ratio and H_2 yields [4,28]. Moreover, there are several studies reporting that H_2 yields were inversely proportional to the butyrate/acetate ratio [38,50,51]. A possible explanation for these contradictory results could be related to acetic acid and butyric acid producing pathways. Theoretically, 4 and 2 mol H_2 /mol glucose can be obtained when H_2 is produced through acetic acid and butyric acid production pathways, respectively (Eq. (3) and Eq. (4)).



However, higher acetic acid productions can also be caused by H_2 consumption to acetic acid production by homoacetogens (Eq. (5)).



Therefore, higher acetate concentrations during DF may not always be accompanied by higher H_2 yields.

VFAs generation resulted in a pH drop at the end of the first stage for all treatments. Table 3 shows the pH values at the end of the first stage for the different pre-treatments, ranging from 4.23 (WFO) to 4.69 (alkaline). The lowest pH value for

Table 2 – Modified Gompertz equation coefficients for different inoculum pre-treatments.

Inoculum pre-treatment	R_m (mL H_2 /h)	λ (h)	P (ml)	R^2
First stage				
Waste frying oil	4.0	5.8	110.6	0.9864
Alkaline	4.1	3.0	61.2	0.9991
Aeration	2.5	3.2	51.3	0.9985
Heat shock	3.3	12.4	81.9	0.9853
Control	3.7	2.4	39.1	0.9996
Second stage				
	R_m (mL CH_4 /d)	λ (d)	P (ml)	R^2
Waste frying oil	50.0	3.9	903.0	0.9972
Alkaline	41.6	0.9	608.2	0.9955
Aeration	41.9	0.6	534.1	0.9953
Heat shock	51.2	0.7	640.0	0.9945
Control	33.4	0.3	438.8	0.9861

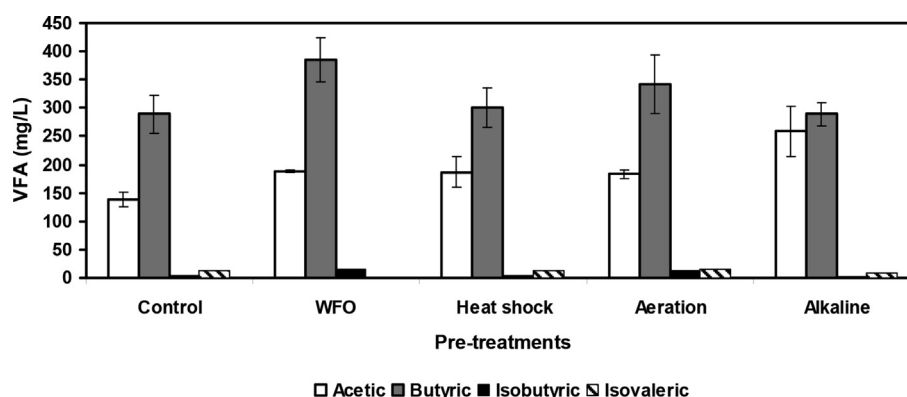


Fig. 2 – VFA concentrations at the end of the first stage of AD.

Table 3 – pH values at the end of the first and second stage for different pre-treatment methods.

	Control	Heat shock	WFO	Alkaline	Aeration
First stage	4.56 ± 0.19	4.42 ± 0.04	4.23 ± 0.15	4.69 ± 0.08	4.30 ± 0.13
Second stage	7.15 ± 0.06	7.21 ± 0.05	7.06 ± 0.02	7.17 ± 0.07	7.14 ± 0.01

WFO pretreated inoculum is in agreement with the highest production of VFAs.

Substrate degradation

The percentages of the initial TOC hydrolysed to DOC have been shown in Table 4. TOC percentages hydrolysed to DOC in the first stage were in the range of 35–38% for all the pre-treatments methods while the hydrolysis performance in the untreated culture was the lowest (28%). The hydrolysis yields reported during DF of FW are in the range of 35–45% [52,53]. During the first stage, no CH₄ was detected in the WFO and heat shock treated cultures while for the aeration and alkali treated cultures, 1.4% and 0.9% of the initial organic carbon was converted to CH₄, respectively. TOC conversion efficiencies for heat shock and WFO were similar (43.4%). TOC conversion for control cultures was less than all treatments (40.58%). The highest TOC conversion during the first stage was obtained for the alkaline pre-treatment (48.6%).

Effect of inoculum pre-treatment methods on the second stage of anaerobic digestion

Methane production

Fig. 3 shows the cumulative CH₄ production during the second stage of AD using the effluents of the DF subjected to the

various inoculum pre-treatments. The CH₄ and CO₂ content in the produced biogas varied between 60–76% and 20–27%, respectively. The highest CH₄ yield of 598.2 mL/gVS was obtained for the WFO pre-treatment. CH₄ yields of 422.91, 432.2 and 381.6 were obtained for heat shock, alkaline, and aeration pre-treatment, respectively. Similar to the first stage, the lowest CH₄ yield was obtained for the untreated cultures (321.5 mL CH₄/gVS). The significant differences in the CH₄ yields for the cultures undergoing different pre-treatments in the first stage (while they were received the same amount of raw granular sludge for the second stage) suggests that the first stage can also play an important role on the efficiency of the second stage. Firstly, the inoculum pretreatment used in the first stage affects the amount of VFAs produced which are then converted to CH₄ by acetoclastic methanogens in the second stage. Second, the inoculum pre-treatment not only has an influence on methanogens but also on many other species present in the mixed culture. This effect may change the behaviour of the granular sludge in terms of many characteristics including hydrolysis.

The modified Gompertz model was used to evaluate CH₄ production performance in the second stage of AD. Kinetic parameters calculated using the modified Gompertz equation for the second stage of AD are shown in Table 2. Regarding the cumulative CH₄ production potential (P), the highest value

Table 4 – Average percentages of initial organic carbon (TOC) degraded during two-stage anaerobic digestion.

Inoculum Pre-treatment	First stage			Second stage		Overall TOC removal (%)
	Hydrolysed to DOC (%)	Emitted as CO ₂ (%)	Emitted as CH ₄ (%)	Emitted as CO ₂ (%)	Emitted as CH ₄ (%)	
Heat shock	35.22 ± 3.34	8.25 ± 0.96	0.00 ± 0.00	14.56 ± 0.78	51.64 ± 1.45	74.45
Waste frying oil	35.04 ± 3.55	8.41 ± 0.52	0.00 ± 0.00	23.15 ± 1.32	64.21 ± 1.61	95.77
Aeration	36.23 ± 4.47	8.87 ± 0.87	1.40 ± 0.03	18.03 ± 0.96	42.76 ± 1.21	71.06
Alkaline	37.98 ± 0.77	9.81 ± 0.59	0.90 ± 0.01	19.08 ± 1.35	48.52 ± 1.11	78.31
Control	28.85 ± 2.77	9.73 ± 0.04	2.01 ± 0.04	15.03 ± 0.61	36.14 ± 2.92	62.91

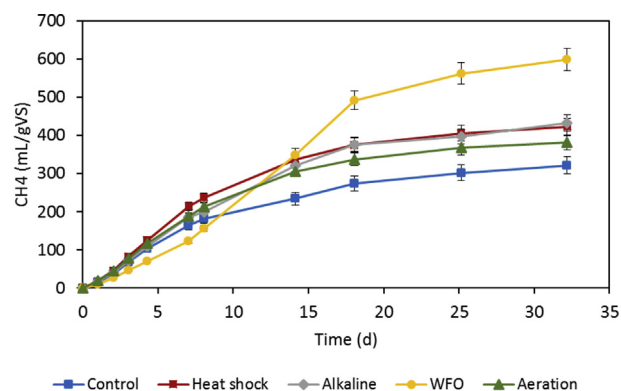


Fig. 3 – Cumulative CH₄ production for various inoculum pre-treatment methods.

was obtained for the inoculum pretreated with WFO (903 mL) which was 106% higher than the untreated cultures (438.8 mL). The lowest CH₄ production potential among the four pre-treatment methods was recorded for aeration pretreated inoculum (534.1 mL) but was still higher compared to the untreated inoculum.

As for the lag time (λ), the longest value of 3.9 d was obtained for the WFO pretreatment. A possible explanation for this observation could be the partial loss of acetogens (which convert VFAs to acetate) during the first stage of AD. The inhibitory effect of LCFAs on acetogens has been proven by several studies [54,55]. The highest CH₄ production rates (R_m) of 51.2 and 50.2 mL CH₄/d were obtained for heat shock and WFO, respectively. CH₄ production rates of 41.9, 41.6, and 33.4 d were calculated for aeration, alkaline, and control pre-treatments, respectively.

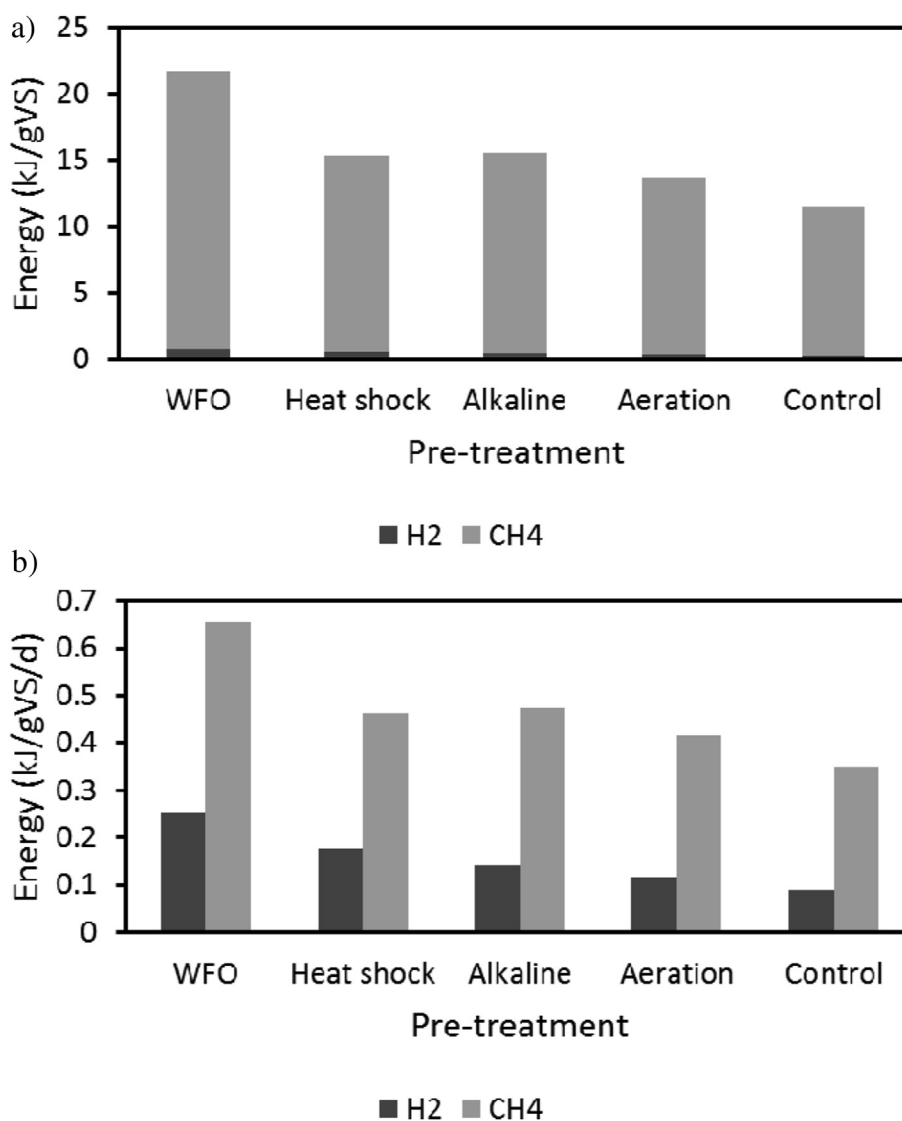


Fig. 4 – a) Total energy recovery from H₂ and CH₄ and b) Conversion rates for H₂ and CH₄ using different inoculum pre-treatment methods.

Table 5 – Two-stage AD of organic wastes using various inoculum pre-treatment methods.

Substrate	Inoculum pre-treatment method	H ₂ yield (mL/gVS)	CH ₄ yield (mL/gVS)	Total energy yield (kJ/gVS)	Reference
Household waste	Heat shock (100 °C, 1 h)	43.0	511.0	18.31	[59]
Vinegar residues	Alkaline (NaOH, pH 12, 24 h)	53.2	192.0	7.25	[6]
Potato waste	Heat shock (70 °C, 2 h)	85.0	364.0	13.59	[13]
Food waste	Heat shock (90 °C, 30 min)	55.0	94.0	3.84	[60]
Food waste	Heat shock (105 °C, 4 h)	104.1	99.0	4.50	[61]
Food waste	Heat shock (100 °C, 30 min)	43.0	511.6	18.32	[14]
Food waste	Heat shock (120 °C, 20 min)	11.8	391.2	14.10	[52]
Food waste	WFO (24 h, 15 g/L and then washed)	76.1	598.2	21.69	This study
Food waste	Heat shock (90 °C, 30 min)	53.8	422.9	15.34	This study

The pH values at the end of the second stage were in the range of 7.06 (WFO) to 7.21 (heat shock), which are close to the optimum pH for methanogens [56] (Table 3).

TOC removal

During DF, a major fraction of the initial TOC remains in the liquid phase in the form of organic acids that can be further converted to CH₄. Coupling DF and AD could enhance the substrate conversion and subsequently result in a more stabilized digestate that can be used as a fertilizer [8,57].

As shown in Table 4, during the first stage of AD, carbon removal was in the range of 8–11% of the initial TOC (in the form of CO₂ or CH₄). This removal means that approximately 90% of the initial TOC remained in the solid or liquid phases at the beginning of the second stage of AD. The lowest carbon removal during the second stage was observed for the untreated cultures (51.17% of the initial TOC). The lowest carbon conversion to CH₄ for the untreated cultures could be attributed to the lower conversion of substrate to organic acids in the first stage. The highest carbon removal was obtained by WFO with 87.36% of the initial TOC removal during the second stage. The rank of TOC removal relative to other inoculum pre-treatments was alkaline > heat > aeration.

Overall performance of the two-stage process

Energy conversion efficiency of the substrate in DF is low due to the major conversion of substrate to soluble metabolites. Combining DF with AD could improve the overall energy yield of the system and enhance the substrate degradation. In order to analyse the overall performance of the two-stage AD, carbon removal and total energy yield from H₂ and CH₄ were evaluated for all of the inoculum pre-treatments as well as untreated cultures.

The overall energy yield from two-stage AD was calculated as the energy obtained from H₂ in the first stage together with energy yield from CH₄ in the second stage of AD. Fig. 4a shows the overall energy yield of the two-stage process using different inoculum pre-treatments. The energy yields obtained from H₂ and CH₄ were calculated based on their corresponding calorific values (142 and 55 kJ/g, respectively) [52]. Two-stage AD using inoculum pretreated with WFO resulted in the highest total energy yield of 21.69 kJ/gVS, which was significantly higher than the untreated cultures (11.52 kJ/gVS). Total energy yields of 15.34, 15.55, and 13.68 kJ/gVS were obtained for heat shock, alkaline, and aeration, respectively.

Moreover, the conversion rates of H₂ and CH₄ for all of the pre-treatments were higher relative to the control (Fig. 4b).

Total TOC removal was calculated as the sum of TOC emitted as CO₂ and CH₄ during two-stage AD. The two-stage process resulted in an acceptable carbon removal (95.77% initial TOC) for WFO pre-treatment. The range for the other inoculum pre-treatments was between 71.06% and 78.31%. The lowest carbon removal for the untreated cultures (62.91%) might indicate that there was a positive effect of inoculum pre-treatment on substrate degradation. Rodríguez-Méndez et al. [58], reported that elevated levels of LCFAs are not inhibitory for hydrolysing and acidogenic bacteria. This means that LCFAs could be used to selectively enrich H₂ producing bacteria without reducing the hydrolysis efficiencies.

Table 5 shows the H₂ and CH₄ yields obtained from two-stage AD of organic substrates in the literature. Additionally, the total energy yields obtained by two-stage processes have been shown to be compared with the results of the present study. As previously mentioned, the majority of the two-stage AD studies have used heat shock to enrich H₂ producing bacteria in the first stage. The total energy recovery in terms of H₂ and CH₄ obtained in the present study is among the highest values found in the literature. However, it should be mentioned that the composition variability of the FW as well as the composition of the inoculum can affect the H₂ and CH₄ yields. Therefore, any direct comparisons should be made with caution. In the present study, which was performed with similar substrates and inoculum for all of the pre-treatments, the total energy yield obtained by WFO pre-treatment was higher than the other pre-treatments studied.

Conclusion

Two-stage AD tests were performed to measure the H₂ and CH₄ recovery from FW. The effect of the four inoculum pre-treatment methods namely heat shock, aeration, and alkaline, pre-treatment with WFO to enrich H₂ producing bacteria in the first stage was studied. Moreover, the possible effect of the inoculum pre-treatment method on the second stage was also evaluated. This study revealed that inoculum pre-treatment not only could increase H₂ yields but it also has an impact on the second stage of AD and CH₄ production. Inoculum pre-treatment with WFO resulted in the highest H₂ and CH₄ yield in the first and second stages, respectively. The highest total energy yield and substrate degradation

efficiencies were obtained using inoculum pre-treatment with WFO. In order to better analyse the effect of inoculum pre-treatment methods on enhanced H_2 and CH_4 yields, further studies should be performed to investigate the microbial community changes during the first and second stages of AD. Energetic costs due to inoculum pretreatment would need also to be considered in order to calculate the net energetic production.

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